PDGuard: A CNN-BiLSTM Attention Network for Wearable IMU-Based Parkinson's Disease Detection

***Abstract*—Parkinson's disease (PD) is commonly assessed through specialist examination, which limits frequent monitoring outside clinical settings. We present PDGuard, a compact network that combines 1D convolution, bidirectional LSTM layers, and multi-head self-attention to classify raw wrist-worn accelerometer and gyroscope signals. Evaluation uses the public PADS cohort (469 participants) with strictly subject-disjoint five-fold cross-validation. On PD versus healthy controls, PDGuard achieves 88.34% accuracy (±1.01), 83.79% balanced accuracy (±1.13), a mean fold ROC-AUC of 0.915, and a pooled held-out ROC-AUC of 0.918. On the harder PD versus differential-diagnosis task, it achieves 77.34% accuracy and 72.89% balanced accuracy. All baselines use the same folds, windows, and preprocessing. The model has 1.24M parameters and requires 18.3 ms per 4.8 s window on the tested Colab CPU, supporting further evaluation for smartphone-class deployment. Source code will be made publicly available upon publication.**

***Index Terms*—Parkinson's Disease, Inertial Measurement Unit, Convolutional Neural Network, Bidirectional LSTM, Attention Mechanism**

I. Introduction

Parkinson's Disease (PD) is the second most common neurodegenerative disorder, affecting more than 10 million people worldwide with the figure projected to double by 2040 [1]. It arises from the loss of dopaminergic neurons in the substantia nigra, producing motor signs such as resting tremor, rigidity, bradykinesia, and postural instability, and is preceded by a prodromal period of several years, which makes early detection valuable. Clinical diagnosis relies on the UPDRS and Hoehn–Yahr scales [2]; these assessments are time-consuming, require trained clinicians, and remain subject to inter-rater variability. In regions with few movement-disorder specialists, access to repeated assessment is limited.

Cheap MEMS IMUs in wrist-worn devices capture tri-axial acceleration and angular velocity and offer an objective, low-cost route to repeated motor assessment. Wearable-sensor reviews identify tremor, gait, motor fluctuations, and home monitoring as major application areas, while also stressing the need for clinically interpretable and externally validated systems [3]–[5]. Smartphone studies have likewise demonstrated remote symptom measurement and severity estimation [6]–[9]. Extracting reliable information from raw, noisy IMU streams nevertheless remains challenging. A further difficulty is that PD is not the only source of these motor signs: essential tremor, atypical parkinsonism, and secondary parkinsonism overlap with idiopathic PD precisely where clinical diagnosis is least reliable. A system that only separates PD from healthy subjects therefore addresses the easier half of the problem; we evaluate both screening and differential-diagnosis settings.

Earlier work relied on handcrafted time- and frequency-domain features fed to classical classifiers such as SVM and Random Forests; these depend heavily on domain knowledge and transfer poorly across subjects. More recent deep models based on 1D-CNN or vanilla LSTM improve on this but individually struggle to capture both short-window local patterns and long-range temporal dependencies. We therefore present PDGuard, an end-to-end model combining stacked 1D-CNN feature extraction, bidirectional LSTM temporal modelling, and multi-head self-attention that re-weights the informative parts of the signal.

Our contributions are: (1) PDGuard, a CNN-BiLSTM-attention framework operating on raw wrist IMU signals with no manual feature engineering; (2) a strictly subject-disjoint five-fold evaluation on PADS in which every baseline is re-implemented under the identical protocol, so all comparisons are controlled rather than quoted across datasets; (3) results on the clinically harder PD-versus-differential-diagnosis task, not only PD-versus-healthy-control; (4) ablation and sensor-modality studies covering accelerometer, gyroscope, and single- versus bilateral-wrist placement; and (5) an analysis of attention weights against signal spectral content together with parameter count and measured inference latency for edge deployment.

II. Related Work

Classical and deep wearable models. Early PD-detection systems used manually engineered time- and frequency-domain descriptors with classifiers such as SVM and Random Forests. On PADS, Varghese *et al.* [11] established baselines from statistical and spectral features. Such pipelines require sensor-specific feature design and may transfer poorly across subjects. Deep approaches increasingly learn representations from raw or transformed sensor signals: recurrent models have been used for wearable motor-state estimation [13], while CNN and time–frequency models are common in freezing-of-gait detection [14], [15]. A recent hybrid CNN–BiLSTM attention model further illustrates the move toward end-to-end wearable analysis [16]. Because cohorts, placements, tasks, and labels differ across studies, we do not compare their published scores directly; instead, representative architecture families are re-implemented on PADS under one protocol.

Placement and attention. Bilateral recordings can add information but increase device burden. Brenner *et al.* [17] showed that a reduced one-sided PADS examination can preserve much of the classification performance, motivating our explicit single- versus bilateral-wrist study. Attention is also emerging in wearable PD analysis. For PADS, Zannat [18] proposed a self-supervised cross-attention encoder for bilateral wrist IMU signals. PDGuard instead retains a recurrent backbone: the BiLSTM models temporal order, while attention re-weights the intervals most useful for classification. The contribution is therefore a controlled, subject-disjoint comparison of standard model families rather than a cross-paper numerical claim.

III. Dataset Description

A. PADS: Parkinson's Disease Smartwatch Dataset

We use the publicly available PADS dataset [10], [11], distributed through PhysioNet [12] under CC BY-NC-SA 4.0. It was collected (2018–2021) at University Hospital Münster: seated participants wore an Apple Watch Series 4 on each wrist while performing 11 neurologist-designed movement tasks spanning resting, postural, and kinetic categories (e.g. resting with eyes closed, resting under cognitive load, holding a 1 kg weight, drinking, nose-touching). Each watch records six-axis inertial data per wrist (three accelerometer axes in g, three gyroscope axes in rad/s) at 100 Hz.

The cohort comprises 469 participants: 276 PD patients, 79 age-matched healthy controls (HC), and 114 patients with differential diagnoses (DD) that mimic PD, including essential tremor, atypical and secondary parkinsonism, and multiple sclerosis. All diagnoses are ICD-10 coded and clinician-confirmed. The dataset holds 5,159 measurement steps recorded bilaterally (10,318 single-wrist records), each of 1,024 samples (10.24 s at 100 Hz). We use only the inertial signal, not the accompanying questionnaire, to establish what is recoverable from the IMU alone. Three properties make PADS suitable here: its sensor modality, placement, and sampling rate match our target deployment; its DD group allows the harder differential task to be posed directly; and its diagnoses are clinician-assigned. We did not adopt the larger mPower cohort, whose signals come from a handheld smartphone with self-reported labels.

B. Task Definitions

We pose two binary tasks. **Task A (PD vs. HC)** compares 276 PD against 79 HC (355 subjects, class ratio ≈ 3.49:1); this is the conventional screening benchmark, for which we report balanced accuracy, sensitivity, specificity, and AUC-ROC alongside accuracy. **Task B (PD vs. DD)** compares 276 PD against 114 DD (390 subjects), the harder differential setting in which both groups present with movement abnormality.

C. Data Preprocessing

Preprocessing is identical for both tasks. The distributed records already have the first 48 onset samples (vibration cue) removed, leaving 976 usable samples per channel; the six axes of one wrist form one input instance (left and right treated as separate instances sharing a subject label, with a 12-channel bilateral variant in Section V-D). A fourth-order zero-phase Butterworth bandpass (0.5–40 Hz) removes the gravitational offset and high-frequency noise while preserving the tremor (3–7 Hz) and voluntary-movement (0.5–3 Hz) bands; no resampling is applied. Each record yields windows of $T = 480$ samples (4.8 s): at training time a single random temporal crop provides continuous augmentation, and at evaluation five overlapping windows per record are averaged (dense test-time windowing), never sharing a window across the train/test boundary. Per-channel z-score normalisation is computed within each window, which removes inter-subject amplitude variance, equalises accelerometer and gyroscope units, and prevents leakage across splits. Following the dataset authors' recommendation, we weight the binary cross-entropy loss inversely to training-fold class frequency and report balanced accuracy as the headline metric; we avoid synthetic oversampling (e.g. SMOTE), which produces physically meaningless inertial signals and would break the subject-disjoint guarantee below. After windowing, Task A yields 7,810 wrist records (39,050 evaluation windows) and Task B yields 8,580 (42,900).

D. Evaluation Protocol

All results use **five-fold subject-disjoint cross-validation**. PADS documents that recommended five-fold splits exist but distributes no literal fold-assignment file [11], so we use a locally-seeded subject-wise stratified group K-fold (scikit-learn `StratifiedGroupKFold`, `random_state=42`) as a documented substitute, applied independently per task. Every window, task, and both wrists of a subject fall in one fold; because each subject contributes 11 tasks $\times$ 2 wrists, a record-level split would leak near-duplicate windows across the boundary and inflate scores. Within each training fold, 15% of *subjects* is held out for early stopping and LR scheduling. Unless stated otherwise, reported values are means over the five test folds; standard deviations are included where available. **Window level** metrics treat each 4.8 s window independently; **subject level** metrics average all of a subject's window probabilities (across all tasks and both wrists) with a validation-tuned threshold, the granularity at which a screening decision would actually be made.

IV. Proposed Methodology

A. PDGuard Architecture Overview

PDGuard is an end-to-end network that maps a raw 6-channel IMU time series to a PD versus non-PD decision with a confidence score. As shown in Fig. 1, it has four stages: input representation, convolutional feature extraction, bidirectional temporal modelling, and attention-weighted classification.

B. Convolutional Feature Extraction

For input $\mathbf{X} \in \mathbb{R}^{T\times C}$ ($T = 480$, $C = 6$: 3-axis accelerometer + 3-axis gyroscope), three stacked 1D-CNN blocks extract multi-scale temporal features with filters increasing 64/128/256 and kernels decreasing 7/5/3. Each block ends in size-2 max pooling, reducing the sequence eightfold from $T = 480$ to $T' = 60$; this shortening keeps both the recurrent and quadratic-attention costs tractable to train and deploy. The l-th block is

$$\mathbf{H}^{(l)} = \mathrm{MaxPool}_2\Big(\mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathbf{W}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}^{(l)}\Big)\Big)\Big) \tag{1}$$

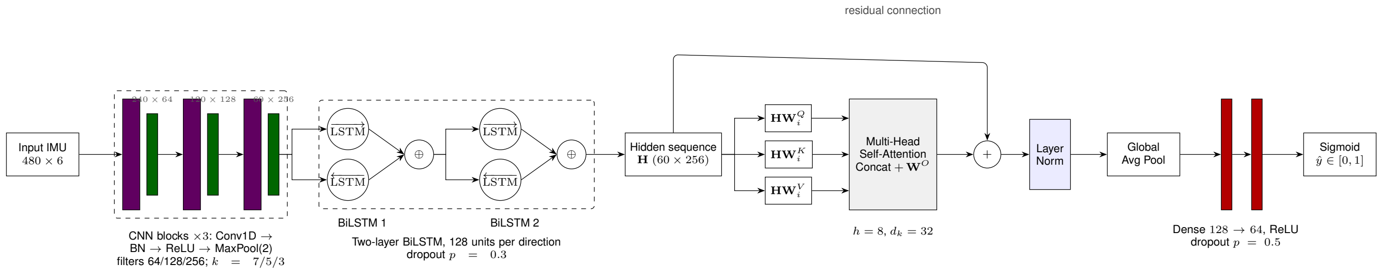


Fig. 1: PDGuard architecture: three CNN blocks, a two-layer BiLSTM, eight-head self-attention, global pooling, and classification.

where $*$ is 1D convolution with unit stride and *same* padding, and BN is batch normalisation.

C. Bidirectional LSTM Temporal Modelling

The CNN output $\mathbf{H}^{(3)} \in \mathbb{R}^{T' \times 256}$ feeds a 2-layer bidirectional LSTM with 128 units per direction (256-dim output per timestep). Processing the sequence in both directions captures dependencies from past and future context, which suits the oscillatory nature of PD tremor and the periodicity of the repetitive tasks. At timestep t,

$$\mathbf{h}_t = \left[\overrightarrow{\mathbf{h}_t} \oplus \overleftarrow{\mathbf{h}_t}\right] \tag{2}$$

with $\oplus$ concatenation; dropout ($p = 0.3$) follows each LSTM layer.

D. Multi-Head Self-Attention

The hidden-state sequence $\mathbf{H} \in \mathbb{R}^{T' \times d_{\text{model}}}$ ($d_{\text{model}} = 256$) enters a multi-head self-attention (MHSA) layer, letting the network emphasise informative intervals (e.g. sustained tremor) rather than weighting all timesteps equally. Each of $h = 8$ heads computes

$$\text{head}_i = \text{Attention}\left(\mathbf{H}\mathbf{W}_i^Q, \mathbf{H}\mathbf{W}_i^K, \mathbf{H}\mathbf{W}_i^V\right) \tag{3}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{4}$$

with $\mathbf{W}_i^{Q,K,V} \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $d_k = d_{\text{model}}/h = 32$; the $\sqrt{d_k}$ scaling prevents the softmax from saturating. The heads are concatenated and projected,

$$\text{MHSA}(\mathbf{H}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O \tag{5}$$

$\mathbf{W}^O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, and wrapped in a residual connection and layer normalisation, $\mathbf{Z} = \text{LayerNorm}(\mathbf{H} + \text{MHSA}(\mathbf{H}))$, which we found necessary for stable convergence. We omit positional encoding, as the BiLSTM already yields order-dependent states.

E. Classification Head

Global average pooling over $\mathbf{Z}$ gives a 256-dim context vector, passed through two dense layers (128, 64; ReLU; dropout $p = 0.5$) to a sigmoid output $\hat{y} \in [0, 1]$ (window-level threshold 0.5; subject-level thresholds are selected on validation data). We optimise the class-weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{6}$$

where w_{y_i} is the inverse training-fold frequency of class y_i.

F. Training Configuration

The TensorFlow 2.12 implementation uses AdamW (initial learning rate 10^{-3}, weight decay 10^{-4}), batch size 64, and at most 100 epochs. The learning rate is halved after five epochs without validation improvement; early stopping follows after 15 epochs and restores the best weights. CNN filters are 64/128/256 with kernels 7/5/3, the two BiLSTM layers use 128 units per direction, and dropout is 0.3 in the recurrent stack and 0.5 in the classifier. Training used a single NVIDIA T4 GPU in Google Colab.

G. Baseline Implementations

All baselines use the same folds, windows, and preprocessing as PDGuard; they represent standard wearable-PD model families rather than exact reproductions of individual publications. SVM (RBF) and Random Forest use a 60-dimensional per-window vector containing, for each channel, mean, standard deviation, skewness, kurtosis, RMS, zero-crossing rate, dominant frequency, spectral entropy, and band power at 0.5–3 and 3–7 Hz. The 1D-CNN uses PDGuard's convolutional trunk and classifier, CNN-LSTM adds a two-layer unidirectional LSTM, and CNN+STFT applies a 2D-CNN to per-channel STFT spectrograms.

V. Experimental Results

A. Evaluation Metrics

We report accuracy (Acc), balanced accuracy (BAcc), sensitivity (Sens), specificity (Spec), F1, and ROC-AUC. BAcc and AUC are emphasised because Task A is imbalanced. In Table I, Acc and BAcc include the available fold mean ± standard deviation; the remaining entries are fold means. Figure 2 instead pools the held-out window predictions from all five folds, so its AUC values can differ slightly from the fold-mean AUCs in the table.

TABLE I: Task A results. Acc/BAcc are mean ± SD across five folds.

Method	Acc	BAcc	Sens	Spec	F1	AUC
Window level						
SVM (RBF)	78.90±0.89	72.18±1.20	84.31	60.06	86.13	0.786
Random Forest	81.82±0.87	76.21±1.36	86.31	66.10	88.07	0.837
1D-CNN	85.82±0.99	80.49±0.84	90.10	70.89	90.81	0.870
CNN-LSTM	85.52±1.20	80.62±1.98	89.44	71.80	90.57	0.894
CNN+STFT	85.50±1.03	80.70±1.39	89.35	72.04	90.55	0.869
PDGuard	**88.34±1.01**	**83.79±1.13**	**92.00**	**75.58**	**92.46**	**0.915**
Subject level						
CNN-LSTM	89.29±1.30	86.82±0.99	91.30	82.33	–	–
PDGuard	**92.11±1.64**	**89.48±1.81**	**94.21**	**84.75**	–	**0.957**

TABLE II: Task B window-level results (PD vs. DD).

Method	Acc	BAcc	Sens	Spec	AUC
SVM (RBF)	65.89	60.30	73.75	46.86	0.634
Random Forest	70.10	65.04	77.23	52.84	0.676
1D-CNN	71.90	66.86	78.99	54.73	0.706
CNN+STFT	74.64	69.96	81.21	58.71	0.724
CNN-LSTM	74.46	70.12	80.57	59.67	0.747
PDGuard	**77.34**	**72.89**	**83.59**	**62.19**	**0.771**

B. Task A: PD versus Healthy Controls

Table I gives window- and subject-level performance under the same five subject-disjoint folds.

The classical baselines perform worst, while the three deep baselines are tightly grouped. Their window-level BAcc values differ by only 0.21 points, less than their fold-to-fold variation. PDGuard exceeds the strongest of this group (CNN+STFT) by 3.09 points. The root-sum-square of the two reported standard deviations is 1.79 points, so the observed margin is larger than this descriptive spread, although this comparison is not a formal significance test. At subject level, averaging window probabilities improves both reported models. PDGuard's BAcc lead over CNN-LSTM is 2.66 points, compared with a combined standard deviation of 2.06. The subject-level AUC of 0.957 is a separate pooled quantity and should not be compared directly with a mean of fold-specific ROC curves.

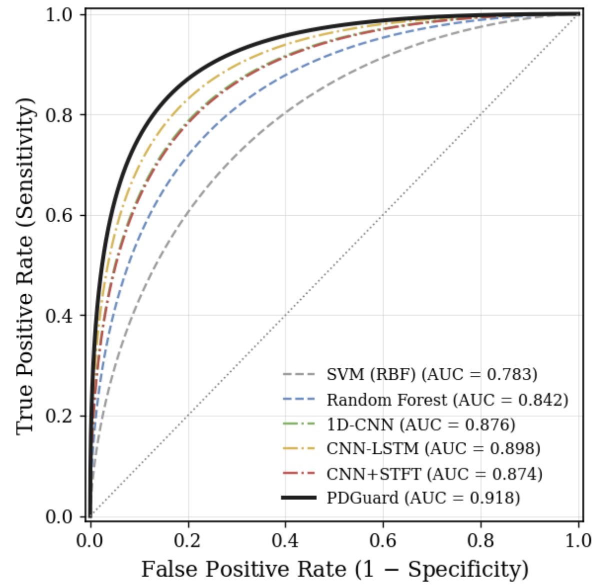
C. Task B: PD versus Differential Diagnoses

Task B replaces the 79 healthy controls with all 114 differential-diagnosis subjects. Both classes therefore contain abnormal movement, making this a differential rather than a screening task. Table II reports the supplied window-level fold means.

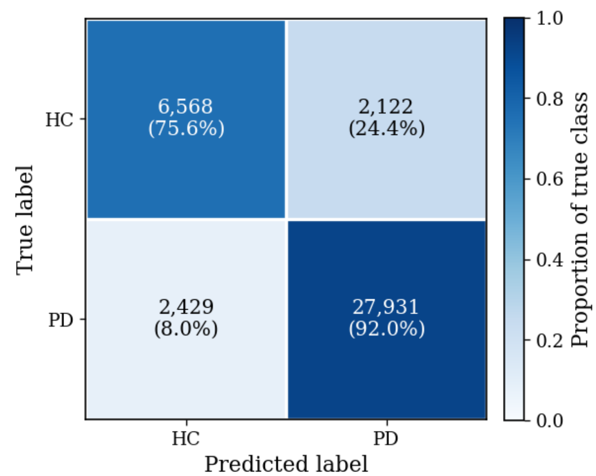
PDGuard's BAcc decreases from 83.79% on Task A to 72.89% on Task B, while CNN-LSTM falls from 80.62% to 70.12%. Most of the reduction is in specificity: the negative class now contains disorders with overlapping motor signs, so the classifier must distinguish the character of parkinsonian movement rather than merely detect abnormal movement. The lower performance therefore reflects the intended difficulty of the differential task.

D. Ablation and Sensor Studies

To keep the compute budget tractable, the ablation and sensor studies run on a single fixed fold (fold 1); absolute



(a) Pooled window-level ROC curves.



(b) Pooled window-level confusion matrix.

Fig. 2: Task A pooled window-level predictions across the five held-out folds.

values are not directly comparable with the five-fold tables above, but the relative ordering is. Table III reports both: the top panel ablates architectural components, the bottom panel varies sensor modality and wrist placement.

Adding a unidirectional LSTM to the CNN gives the largest accuracy gain (+1.57), and making the recurrent layer bidirectional adds a further 1.37 points. A single attention head adds 0.19 points and the full eight-head layer adds another 0.32. Because this study uses one fixed fold, these small attention differences should be treated as suggestive rather than statistically resolved. For sensing, the accelerometer alone outperforms the gyroscope, while the six-channel IMU outperforms either modality alone. Bilateral input improves accuracy from 88.31% to 89.36%; this one-point gain suggests that a single-watch implementation is a reasonable trade-off.

TABLE III: Task A ablation and sensor study (single fold).

Configuration	Acc	F1/Sens	AUC	Spec/Δ
Component ablation (F1, ΔAcc)				
CNN only	84.81	90.14	0.860	-3.45
CNN + LSTM (unidir.)	86.38	91.19	0.886	-1.88
CNN + BiLSTM (no attn.)	87.75	92.08	0.898	-0.51
CNN + BiLSTM + 1-head attn	87.94	92.19	0.905	-0.32
PDGuard (full)	**88.26**	**92.41**	**0.913**	–
Sensor modality / placement (Sens, Spec)				
Accelerometer only (3 ch)	86.49	90.93	0.877	70.97
Gyroscope only (3 ch)	83.56	88.23	0.849	67.22
Single wrist, IMU (6 ch)	88.31	91.95	0.910	75.57
Bilateral, IMU (12 ch)	89.36	92.97	0.922	76.70

TABLE IV: Measured model complexity and CPU latency.

Model	Params (M)	Size (MB)	Latency (ms)	BAcc
1D-CNN	0.18	0.73	3.1 ± 0.19	80.49
CNN-LSTM	0.50	1.99	11.4 ± 0.71	80.62
CNN+STFT	0.91	3.65	6.8 ± 0.42	80.70
PDGuard	**1.24**	**4.94**	$\mathbf{18.3 \pm 1.14}$	**83.79**

TABLE V: Descriptive 4–6 Hz power in high- versus low-attention intervals.

Group	Power ratio	Effect size r	Windows
PD	2.34	0.64	30,360
DD	1.62	0.25	12,540
HC	1.18	0.06	8,690

E. Model Complexity and Deployment

Latency was measured per 4.8 s window with batch size 1, averaged over 100 runs on the Colab CPU. These are host-CPU measurements, not target-device benchmarks.

PDGuard contains 1,235,073 parameters and averages 18.3 ms per window on the tested CPU. Since each input spans 4.8 s, this throughput is compatible with real-time processing, but latency must still be measured on the intended phone or single-board computer. Quantised size and latency are not reported because the actual TFLite export has not yet established whether the recurrent kernels remain in fp32. Microcontroller deployment is not claimed.

VI. Discussion

A. Attention–Spectrum Association

Attention weights are temporal, not spectral. We therefore test whether intervals with high attention have different 4–6 Hz power from intervals with low attention. The eight attention maps are averaged into a per-timestep score and the highest and lowest quartiles are compared.

High-attention intervals in the PD group contain 2.34 times the 4–6 Hz power of low-attention intervals, while the HC ratio is close to one and the DD group is intermediate. This pattern is consistent with heterogeneous tremor expression across differential diagnoses. The analysis is descriptive: it identifies an association between temporal attention and spectral content, but does not show that the attention layer directly encodes frequency and does not establish a clinical biomarker.

B. Positioning Relative to Clinical Assessment

The reported sensitivity and specificity measure agreement with the clinical labels in PADS; they are not a comparison with specialist diagnostic accuracy against a longitudinal or neuropathological reference standard. PDGuard should therefore be viewed as a triage or monitoring aid that can support referral, not as a replacement for clinical assessment.

C. Limitations

PADS is a single-site, supervised, seated, task-based dataset; free-living and cross-site performance remain unknown. The 79-person control group makes fold-level specificity coarse, and the cohort's gender distribution may introduce a correlated signal that should be examined with prospectively balanced data. The sample is European, so local validation is required for other populations. The model predicts a binary label rather than severity, and the attention analysis is descriptive rather than evidence of a clinical biomarker. Finally, target-device and quantised-export measurements have not been completed.

VII. Conclusion

PDGuard combines convolutional feature extraction, bidirectional temporal modelling, and multi-head self-attention for PD classification from wrist-worn IMU signals. Under subject-disjoint five-fold evaluation on PADS, it achieves 83.79% BAcc, a mean fold AUC of 0.915, and a pooled held-out AUC of 0.918 for PD versus healthy controls, together with 72.89% BAcc for PD versus differential diagnoses. The controlled baselines show that the recurrent backbone contributes most of the architectural gain, while bilateral sensing adds only about one point over a single wrist. The 1.24M-parameter model runs in 18.3 ms per window on the tested CPU. Future work should evaluate TFLite export on target hardware, control demographic confounding prospectively, and validate the model using free-living and multi-site cohorts.

References

[1] E. R. Dorsey and B. R. Bloem, "The Parkinson pandemic: a call to action," *JAMA Neurology*, vol. 75, no. 1, pp. 9–10, 2018.

[2] C. G. Goetz *et al.*, "MDS-sponsored revision of the Unified Parkinson's Disease Rating Scale," *Movement Disorders*, vol. 23, no. 15, pp. 2129–2170, 2008.

[3] W. Maetzler, J. Domingos, K. Srulijes, J. J. Ferreira, and B. R. Bloem, "Quantitative wearable sensors for objective assessment of Parkinson's disease," *Movement Disorders*, vol. 28, no. 12, pp. 1628–1637, 2013.

[4] E. Rovini, C. Maremmani, and F. Cavallo, "How wearable sensors can support Parkinson's disease diagnosis and treatment: A systematic review," *Frontiers in Neuroscience*, vol. 11, art. 555, 2017.

[5] A. J. Espay *et al.*, "Technology in Parkinson's disease: Challenges and opportunities," *Movement Disorders*, vol. 31, no. 9, pp. 1272–1282, 2016.

[6] S. Arora *et al.*, "Detecting and monitoring the symptoms of Parkinson's disease using smartphones: A pilot study," *Parkinsonism and Related Disorders*, vol. 21, no. 6, pp. 650–653, 2015.

[7] B. M. Bot *et al.*, "The mPower study, Parkinson disease mobile data collected using ResearchKit," *Scientific Data*, vol. 3, art. 160011, 2016.

[8] A. Zhan *et al.*, "Using smartphones and machine learning to quantify Parkinson disease severity: The mobile Parkinson disease score," *JAMA Neurology*, vol. 75, no. 7, pp. 876–880, 2018.

[9] L. Omberg *et al.*, "Remote smartphone monitoring of Parkinson's disease and individual response to therapy," *Nature Biotechnology*, vol. 40, no. 4, pp. 480–487, 2022.

[10] J. Varghese, A. Brenner, L. Plagwitz, C. van Alen, M. Fujarski, and T. Warnecke, "PADS—Parkinson's Disease Smartwatch dataset (version 1.0.0)," *PhysioNet*, 2024, doi: 10.13026/m0w9-zx22.

[11] J. Varghese *et al.*, "Machine learning in the Parkinson's disease smartwatch (PADS) dataset," *npj Parkinson's Disease*, vol. 10, art. 9, 2024.

[12] A. L. Goldberger *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[13] J. Goschenhofer, F. M. J. Pfister, K. A. Yuksel, B. Bischl, U. Fietzek, and J. Thomas, "Wearable-based Parkinson's disease severity monitoring using deep learning," *arXiv preprint* arXiv:1904.10829, 2019.

[14] C. Ahlrichs *et al.*, "Detecting freezing of gait with a tri-axial accelerometer in Parkinson's disease patients," *Medical & Biological Engineering & Computing*, vol. 54, pp. 223–233, 2016.

[15] S. Aich, P. M. Pradhan, J. Park, N. Sethi, V. S. S. Vathsa, and H.-C. Kim, "A validation study of freezing-of-gait detection and machine-learning-based prediction using a wearable accelerometer," *Sensors*, vol. 18, no. 10, art. 3287, 2018.

[16] M. H. Al-Adhaileh *et al.*, "Deep learning techniques for detecting freezing-of-gait episodes in Parkinson's disease using wearable sensors," *Frontiers in Physiology*, vol. 16, art. 1581699, 2025.

[17] A. Brenner, M. Fujarski, T. Warnecke, and J. Varghese, "Reducing a complex two-sided smartwatch examination for Parkinson's disease to an efficient one-sided examination preserving machine learning accuracy," *arXiv preprint* arXiv:2205.05361, 2022.

[18] M. Zannat, "Parkinson's disease detection via self-supervised dual-channel cross-attention on bilateral wrist-worn IMU signals," *arXiv preprint* arXiv:2604.18372, 2026.